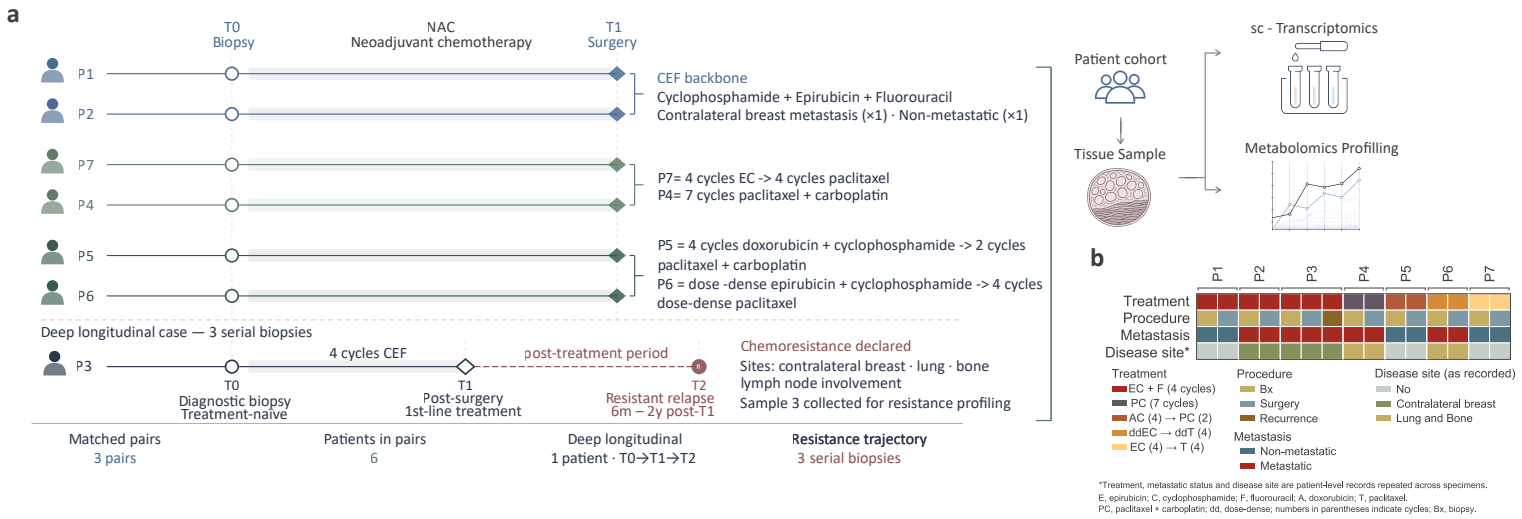

a
T0
Biopsy
NAC
Neoadjuvant chemotherapy
T1
Surgery
P1
P2
P7
P4
P5
P6
CEF backbone
Cyclophosphamide + Epirubicin + Fluorouracil
Contralateral breast metastasis (×1) · Non-metastatic (×1)
P7= 4 cycles EC -> 4 cycles paclitaxel
P4= 7 cycles paclitaxel + carboplatin
P5 = 4 cycles doxorubicin + cyclophosphamide -> 2 cycles paclitaxel + carboplatin
P6 = dose -dense epirubicin + cyclophosphamide -> 4 cycles dose-dense paclitaxel
Deep longitudinal case — 3 serial biopsies
P3
4 cycles CEF
post-treatment period
T0
Diagnostic biopsy
Treatment-naive
T1
Post-surgery
1st-line treatment
T2
Resistant relapse
6m – 2y post-T1
Chemoresistance declared
Sites: contralateral breast · lung · bone
lymph node involvement
Sample 3 collected for resistance profiling
Matched pairs
3 pairs
Patients in pairs
6
Deep longitudinal
1 patient · T0→T1→T2
Resistance trajectory
3 serial biopsies
Patient cohort
Tissue Sample
sc - Transcriptomics
Metabolomics Profilling
b
P1
P2
P3
P4
P5
P6
P7
Treatment
Procedure
Metastasis
Disease site*
Treatment
EC + F (4 cycles)
PC (7 cycles)
AC (4) → PC (2)
ddEC → ddT (4)
EC (4) → T (4)
Procedure
Bx
Surgery
Recurrence
Metastasis
Non-metastatic
Metastatic
Disease site (as recorded)
No
Contralateral breast
Lung and Bone
*Treatment, metastatic status and disease site are patient-level records repeated across specimens.
E, epirubicin; C, cyclophosphamide; F, fluorouracil; A, doxorubicin; T, paclitaxel.
PC, paclitaxel + carboplatin; dd, dose-dense; numbers in parentheses indicate cycles; Bx, biopsy.
c
UMAP2
UMAP1
Epithelial
Endothelial
CAFs
PVL
T-cells
B-cells
Plasmablasts
Myeloid
d
Epithelial
Endothelial
CAFs
PVL
T-cells
B-cells
Plasmablasts
Myeloid
KRT7
TACSTD2
KRT8
KRT19
KRT18
RAMP2
IGFBP7
SPARCL1
CRIP2
RNASE1
COL1A1
COL1A2
DCN
LUM
C1S
MYL9
ACTA2
CALD1
IGFBP7
SPARCL1
BTG1
CD3E
SRGN
CD52
PTPRC
CD37
MS4A1
BTG1
CD52
CD79A
DERL3
IGHG1
MZB1
SSR4
CD79A
AIF1
FCER1G
FTL
TYROBP
CD74
Scaled mean
1.0
0.5
0.0
Expressing
20 40 60 80100
e
UMAP2
UMAP1
Cancer Basal SC
Cancer Cycling
Cancer Her2 SC
Cancer LumA SC
Cancer LumB SC
Luminal Progenitors
Mature Luminal
Myoepithelial
B cells Memory
B cells Naive
Plasmablasts
Cycling_Myeloid
DCs
Macrophage
Monocyte
Endothelial ACKR1
Endothelial CXCL12
Endothelial Lymphatic LYVE1
Endothelial RGS5
CAFs MSC iCAF-like
CAFs myCAF-like
Cycling PVL
PVL Differentiated
PVL Immature
Cycling T-cells
NK cells
NKT cells
T cells CD4+
T cells CD8+
f
UMAP2
UMAP1
CNV score
0.007
0.006
0.004
0.003
0.001
g
CNV score
0.008
0.006
0.004
0.002
0.000
n=27,421
n=41,970
Normal
Tumor
h
Epithelial cells
EPCAM
i
CAF cells
DCN
j
Myeloid cells
C1QA
UMAP2
UMAP1
k
IM
BL1
MSL
M
BL2
LAR
P1 0.28 -0.33 0.11 -0.25 0.01 -0.06
P2 0.10 0.19 -0.22 -0.14 0.22 -0.11
P3 -0.18 -0.01 0.14 0.36 -0.19 -0.12
P4 0.50 -0.12 -0.18 -0.50 0.32 -0.00
P5 -0.49 0.23 0.02 0.51 -0.36 0.04
P6 -0.03 0.23 -0.10 0.02 0.22 0.04
P7 -0.23 -0.15 0.19 0.12 -0.28 0.16
Correlation to subtype centroid
0.6
0.3
0.0
−0.3
−0.6

Figure 4